Explicit infinity point: checked proof boundary

aburan28/crypto, genus-2 projective-law follow-up, 25 October 2026

Object. For a field K and arbitrary $a_3, a_2, a_1, a_0 \in K$, the explicit infinity chart has ring

$$R = K[z, w]/\big(w^2 - z(1 + a_3 z^2 + a_2 z^3 + a_1 z^4 + a_0 z^5)\big).$$

The checked overlap with the affine chart is the principal open $D(z)$.

Checked result. The complement of $D(z)$ in $\operatorname{Spec} R$ contains exactly one point, the origin ideal (z, w). For any prime $\mathfrak{p}$ containing z, the equation gives $w^2 \in \mathfrak{p}$, and primality gives $w \in \mathfrak{p}$. The pre-existing Lean theorem proves (z, w) is maximal, so $\mathfrak{p} = (z, w)$. Evaluation at $z = w = 0$ proves the origin is outside $D(z)$. This statement is for the explicit affine chart; it is not a claim of projectivity or smoothness of the glued scheme.

Compiler evidence. Pinned Lean v4.34.1 and Mathlib revision

`d13f23b723b8a846827a245b89c10fc7d3f11612`

The command `timeout 180s lake env lean Common22CompletionBridge.lean` exited 0. The four `#print axioms` lines list exactly `propext`, `Classical.choice`, and `Quot.sound`. The adjacent CSV records each theorem, its hypotheses, and status; the SVG shows the dependency and open-obligation graph.

Remaining bridge. A chosen smooth projective completion must be constructed and its two affine charts identified with the present quotient rings, with the same overlap isomorphism. Alternatively, construct a finite map from the glued scheme to $\mathbf{P}^1_K$, then prove the properness and chartwise smoothness hypotheses under an explicit squarefree condition. The pinned Mathlib supplies generic gluing and Proj APIs, but no genus-2 hyperelliptic completion theorem was found. The principal divisor of the interpolating function, including its order at this unique point, and the resulting Pic^0 equation remain separate proof obligations.

Interpretation. This packet advances the local geometry of the constructed chart. It does not revise measured cost ratios or assert correctness of the full published law.